P1286R1
Contra CWG DR1778

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Author:
Richard Smith (Google)

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Abstract

This paper presents a problem with the current resolution of DR1778 and proposes an alternative resolution.

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§ 1. Revision history

Since [P1286R0]:

- Remove discussion of options; EWG has selected their desired alternative.
- Approved unanimously by EWG

§ 2. Background

See lists.isocpp.org/core/2018/01/3741.php for more information.

§ 2.1. CWG 1778: *exception-specification* in explicitly-defaulted functions

Prior to CWG 1778, we required that:

> An explicitly-defaulted function [...] may have an explicit *exception-specification* only if it is compatible with the *exception-specification* on the implicit declaration.

It was observed in LWG 2165 that this creates problems for `std::atomic<T>`, which declares its default constructor thusly:

```
template<typename T> struct atomic {
  atomic() noexcept = default;
```

... which resulted in `atomic<T>` being ill-formed if `T` has a potentially-throwing default constructor.

§ 2.2. Potential fixes

LWG 2165 lists the following as potential fixes:

1. Add nothrow default constructible to requirements for template argument of the generic `atomic<T>`
2. Remove `atomic<T>::atomic()` from the overload set if `T` is not nothrow default constructible.
3. Remove `noexcept` from `atomic<T>::atomic()`, allowing it to be deduced (but the default constructor is intended to be always `noexcept`)
4. Do not default `atomic<T>::atomic()` on its first declaration (but makes the default constructor user-provided and so prevents `atomic<T>` being trivial)
5. A core change to allow the mismatched exception specification if the default constructor isn't used (see c++std-core-21990)

§ 2.3. Language change

CWG chose to resolve the issue by changing the rule to:

> If a function that is explicitly defaulted has an explicit *exception-specification* that is not compatible with the *exception-specification* on the implicit declaration, then
>
> - if the function is explicitly defaulted on its first declaration, it is defined as deleted;
> - otherwise, the program is ill-formed.

That is: implicitly delete the default constructor if the specified exception specification doesn't match the implicit one.

§ 3. Problem

§ 3.1. Existing approach is bad for compilers

Exception specifications are a complete-class context: they are a place where all members of the class *and its enclosing classes* can be used, just like member function bodies, default arguments, and default member initializers. This means we cannot in general determine the implicit exception specification of a member function until we reach the end of the outermost lexically-enclosing class. However, we

need to know which special member functions a class has, and whether or not they are deleted, immediately after the class becomes complete, which (for a nested class) may be earlier.

Example:

```
struct X { X(); };
struct A {
  struct B {
    B() noexcept(A::value) = default;
    X x;
  };
  decltype(B()) b;
  static constexpr bool value = true;
};
A::B b;
```

Here, we do not parse the exception specification for `B::B()` until after we have finished parsing class `A`. But the class `B` becomes complete at its close brace, and at that point we must know the critical facts regarding its definition, including which of its special members are deleted.

Note that we cannot possibly tell whether the call to `B()` within the `decltype` is valid, because we don't know whether `A::B::B()` is deleted yet.

§ 3.2. Existing approach is bad for `std::atomic<T>`

The result of the current wording is that this code is accepted:

```
struct Foo {
  Foo() : n(0) {} // happens to not be noexcept
  int n;
};
std::atomic<Foo> f = Foo(); // ok
```

... but this is ill-formed:

```
std::atomic<Foo> f; // error
```

This appears extremely hard to justify. As LWG 2334 notes,

> Initialization of an atomic object is not an atomic operation.

There appears to be absolutely no reason whatsoever to require the default constructor of `atomic` (or the constructor from a `T`) to be `noexcept`.

§ 3.3. Existing approach prevents a useful feature

Consider the following pattern, which we found several instances of in our codebase when we tightened up the compiler to reject a mismatched exception specification on a defaulted function:

```
struct X {
  std::map<...> m;
  // ... other members
public:
  // I want a defaulted move constructor, and vector<X> needs to be
  // efficient, so please call std::terminate if moving the map throws
  // rather than slowing my code down with unnecessary copies
  X(X &&) noexcept = default;
};
```

Users wanting this feature are forced to write out their own special members, which is an error-prone operation that `= default` was supposed to alleviate.

§ 4. Approach

If the user explicitly specifies an exception specification on a defaulted function, that's the exception specification. Don't delete the function, don't reject the program, just accept it.

Fix `std::atomic<T>` by removing the spurious `noexcept`.

§ 5. Wording

Change in [dcl.fct.def.default]/2:

> The type `T1` of an explicitly defaulted function `F` is allowed to differ from the type `T2` it would have had if it were implicitly declared, as follows:
>
> - `T1` and `T2` may have differing *ref-qualifiers*; and
> - `T1` and `T2` may have differing exception specifications; and
> - if `T2` has a parameter of type `const C&`, the corresponding parameter of `T1` may be of type `C&`.
>
> [...]

Change in [dcl.fct.def.default]/4:

> `~S() noexcept(false) = default;` // ~~deleted: exception specification does not match~~ OK, despite mismatched exception specification

Change in [atomics.types.generic]:

> ```
> template<class T> struct atomic {
>   [...]
>   atomic() noexcept = default;
>   constexpr atomic(T) noexcept ;
> ```

Change before [atomics.types.operations]/2:

> `atomic() noexcept = default;`

Change before [atomics.types.operations]/3:

> `constexpr atomic(T) noexcept ;`

§ References

§ Informative References

[P1286R0]
Richard Smith. Contra CWG DR1778. 5 October 2018. URL: https://wg21.link/p1286r0